

BRAXTON VOGEL

SOFTWARE ENGINEER

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PROFESSIONAL SUMMARY

Software Engineering student at Sam Houston State University (SHSU) with a 3.6 major GPA, pursuing a future M.S. in Computing & Data Science. Strong analytical and problem-solving skills with an applied focus on data analytics, AI-assisted workflows, and technical communication. Proven adaptability through leadership and customer-facing roles.

EDUCATION

Bachelor of Science in Software Engineering; Sam Houston State University (SHSU), Huntsville, Texas

- **General GPA:** 3.4
- **Major GPA:** 3.6
- **Timeframe:** 2024-Present

Certification

- Google Data Analytics Professional Certification (*In progress - June 2026*)
 - SQL Skills
 - Data Visualization Skills
 - Data Cleaning & Transformation Skills
 - Spreadsheet & Basic R Skills

Relevant Pre-College Coursework; Klein Cain High School, Klein, Texas

- Completed both General Academic and Video Game Programming Pathways, developing programming skills through structured coursework and applied projects, which laid the foundation for independent software projects

PROJECT *Additional demos, visuals, and technical breakdowns available in portfolio*

Student Risk Prediction System (AI + ML Web Application) | [GitHub \(braxtonvogel\)](#) Python, Streamlit, Scikik-learn, Ollama

- Built an end-to-end machine learning system that predicts student academic risk using the Open University Learning Analytics Dataset (OULAD)Defined target users including average PC users seeking simplified file organization workflow
- Developed a Random Forest classification model to generate risk probabilities and student-level predictions
- Designed and deployed an interactive Streamlit web application for dataset upload, visualization, and real-time inference
- Integrated a local LLM-powered AI assistant (Nova) using Ollama (Llama 3) for natural language queries and explanation of model outputs
- Implemented automated intervention suggestions to support at-risk students based on model predictions

Command-Line Multi-Client Chat Server | [GitHub \(braxtonvogel\)](#) Java

- Developed a command-line client-server chat application supporting multiple concurrent clients
- Implemented server-side session management with a default lobby and user-created chatrooms
- Designed client interaction flow allowing users to create or join existing chatrooms through menu-based input
- Managed message routing between connected clients to enable real-time text communication
- Practiced structured program design, input validation, and debugging in a networked environment

- Collaborated with several team members to develop a database-driven character management application using Spring Boot, MySQL, and Thymeleaf
- Designed relational database schemas and entity relationships supporting character creation, inventory, spells, and class management workflows
- Performed functional and security testing, including CRUD validation, SQL injection testing, and GUI-to-database consistency checks
- Prepared technical documentation, deployment materials, & GitHub repository organization for final project delivery

Personal Portfolio Website | Live Site (Vercel) | GitHub (braxtonvogel)

HTML, CSS, TypeScript, Next.js

- Designed and developed a responsive personal portfolio website to showcase software engineering projects and technical skills
- Built interactive project display system with dynamic routing and structured project metadata for scalable expansion
- Implemented custom UI components including navigation, animated elements, and an AI-style skill matching widget for engagement
- Integrated GitHub + Vercel CI/CD pipeline enabling automatic deployments on every commit
- Optimized user experience with clean visual hierarchy, responsive layout, and mobile-first design principles

SKILLS

Technical:

- | | |
|---|---|
| • Python, Java, HTML | • Software Engineering: Client-Server Architecture, Basic Networking Concepts, System Design Fundamentals |
| • Machine Learning: Scikit-learn, Random Forest, Data Analysis, Feature Engineering, Data Preprocessing | • Tools & Development: Git, GitHub, Notepad++ |
| • Artificial Intelligence / LLMs: Ollama, Llama 3, Prompt Engineering | • Debugging & Problem Solving: Code Debugging, Error Resolution, Testing and Troubleshooting |
| • Web Development: Streamlit, Spring Boot, Thymeleaf | • Computer Fundamentals: Hardware Fundamentals |

Interpersonal

- Teamwork & Collaboration
- Clear Technical Communication
- Critical Thinking
- Creativity, Problem Solving, and Innovation
- AI-Supported Research, Documentation, and Problem Analysis

Professional:

- AI-Assisted Workflow Optimization
- Data-Driven Decision Making
- Operational Efficiency Management
- Process Improvement and Automation
- Report Generation & Report Writing

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

- **Community Volunteer** – Local church service supporting food distribution for underdeveloped communities
- **Volunteer Participant** – Jeep Club led nonprofit fundraisers (Noah’s ark foundation events, car shows, raffles)
- **Volunteer Support** – Veteran focused initiatives including Wreaths across America, flag placement, & fundraising